Fine-Grained Bird Recognition on CUB-200 with DINOv2 and a Linear Probe

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Abstract

We present a two-stage pipeline for fine-grained bird species classification on CUB-200-2011. Frozen DINOv2 (ViT-B/14) features are extracted once and cached; a logistic regression classifier is then trained on these features in under five minutes. Despite its simplicity, the system reaches ***89.40%*** *Top-1 and* ***98.36%*** *Top-5 test accuracy—competitive with fine-tuned models. We describe the full pipeline, training procedure, and diagnostic visualizations.*

1. Introduction

Fine-grained image recognition—distinguishing visually similar subcategories—is a challenging benchmark for visual features. CUB-200-2011 [1], with 200 North American bird species and ~30 training images per class, is the standard testbed for this problem.

Recent self-supervised vision transformers such as DINOv2 [2] produce rich semantic embeddings that generalize well without task-specific fine-tuning. We exploit this by treating DINOv2 as a fixed feature extractor and applying a lightweight scikit-learn classifier on top, keeping the entire training loop fast and hardware-efficient.

Figure 1. System overview. Feature extraction (Stage 1, GPU) is run once and cached. Classifier training (Stage 2, CPU) and evaluation (Stage 3) are fast and repeatable.

2. Method

2.1 Dataset and Splits CUB-200-2011 provides an official train/test partition. We carve a stratified validation set from training by taking the last 10% of each class's images (sorted by ID), giving reproducible splits without touching the test set.

Split	*Samples*	*Feat. dim*
Train	5,400	768
Validation	594	768
Test	5,794	768

Table 1. Dataset split sizes. All features are L2-normalized float32 vectors from DINOv2 ViT-B/14.

2.2 Stage 1 — Feature Extraction Images are fed through a frozen DINOv2 ViT-B/14 backbone (86M parameters, via HuggingFace). The *[CLS]* token of the final layer is taken as the image representation and L2-normalized before saving. No gradients are computed.

Preprocessing follows the standard ViT protocol: training images use RandomResizedCrop(224, scale 0.5–1.0) + RandomHorizontalFlip; evaluation images use Resize(256) → CenterCrop(224). All splits use ImageNet mean/std normalization.

2.3 Stage 2 — Linear Probe Training A scikit-learn Pipeline of `StandardScaler → LogisticRegression` (multinomial, lbfgs, max_iter=1200) is fitted via GridSearchCV with 5-fold StratifiedKFold. The regularization parameter C is swept over {15, 17.5, 20, 22.5}.

2.4 Stage 3 — Evaluation After training, the serialized model is reloaded to compute Top-1 and Top-5 accuracy on the held-out test set. Four diagnostic figures are generated: (i) top-15 class-confusion heatmap, (ii) t-SNE of 1,500 balanced test samples, (iii) 20 lowest per-class accuracy bars, and (iv) full 200×200 confusion matrix (.npy).

3. Experiments

3.1 Main Results Table 2 reports test accuracy, cross-validation best score, optimal regularization constant, and total training time.

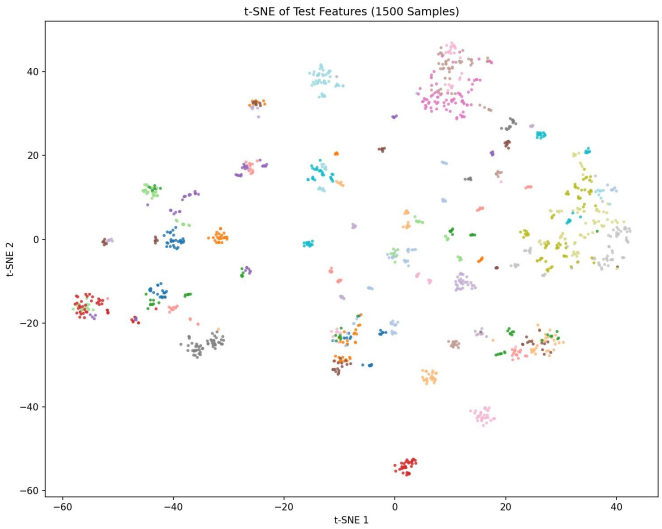

Figure 2. t-SNE of 1,500 balanced test-set DINOv2 features (perplexity=30, PCA init). Each color represents one of the 200 bird species. The clear cluster structure confirms that the frozen features are highly discriminative.

Metric	*Value*
Test Top-1 Accuracy	**89.40%**
Test Top-5 Accuracy	**98.36%**
CV Best Score (val)	88.98%
Best C	17.5
Training Time	~4.7 min

Table 2. Test-set results on CUB-200-2011 (vitb14 backbone).

3.2 Hyperparameter Sweep The C sweep targets a known good neighborhood. CV scores are tightly clustered, confirming the model is insensitive to small changes in regularization at this scale.

C	*Mean CV*	*Std CV*
15.0	88.96%	0.53%
17.5	**88.98%**	0.59%
20.0	88.96%	0.56%
22.5	88.98%	0.55%

Table 3. GridSearchCV results. Bold marks the selected model (C = 17.5).

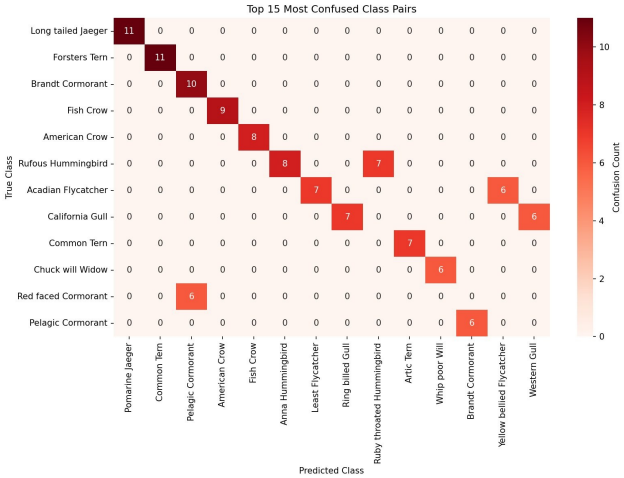

Figure 3. Top-15 most confused class pairs. Dominant confusions occur within ecologically related groups: terns (Forster's vs. Common), jaegers (Long-tailed vs. Pomarine), cormorants (Brandt's vs. Red-faced & Pelagic), crows (American vs. Fish), and hummingbirds (Rufous vs. Anna). These are visually very similar species, suggesting errors are near the limits of fine-grained discriminability.

4. Architecture & Modules

The project comprises three packages. *datasets/* holds the CUB-200 loader and transforms. *models/* contains the DINOv2 extractor. *scripts/* holds feature extraction. Two top-level scripts tie everything together: `train.py` (end-to-end) and `evaluate.py` (metrics and figures).

File	*Role*
`cub200.py`	Dataset class; stratified 90/10 split
`transforms.py`	Train / eval image transforms
`dinov2.py`	Frozen ViT extractor (HuggingFace)
`extract_features.py`	Batch inference → .npz cache
`train.py`	Cache check + GridSearchCV + results.json
`evaluate.py`	Metrics, heatmap, t-SNE, bar chart

Table 4. Module summary. Three DINOv2 variants: vits14 (384-d), vitb14 (768-d, default), vitl14 (1024-d).

5. Usage

5.1 Quick Start The full pipeline runs in two commands:

```
python train.py \
--data_root data/CUB_200_2011 \
--model_variant vitb14 \
--output_dir ./runs/vitb14

python evaluate.py \
--data_root data/CUB_200_2011 \
--run_dir ./runs/vitb14 \
--cache_dir ./cache
```

5.2 Key Arguments

Argument	*Default*	*Note*
`--model_variant`	vitb14	vits14 / vitb14 / vitl14
`--cv_folds`	5	StratifiedKFold folds
`--device`	auto	auto / cuda / cpu
`--tsne_color_mode`	class	class or order
`--tsne_samples`	1500	Balanced t-SNE samples

Table 5. Selected CLI arguments.

5.3 Output Files A completed run produces a serialized classifier, a results JSON file, and a figures subdirectory containing the confusion heatmap, t-SNE plot, per-class accuracy chart, and full confusion matrix array.

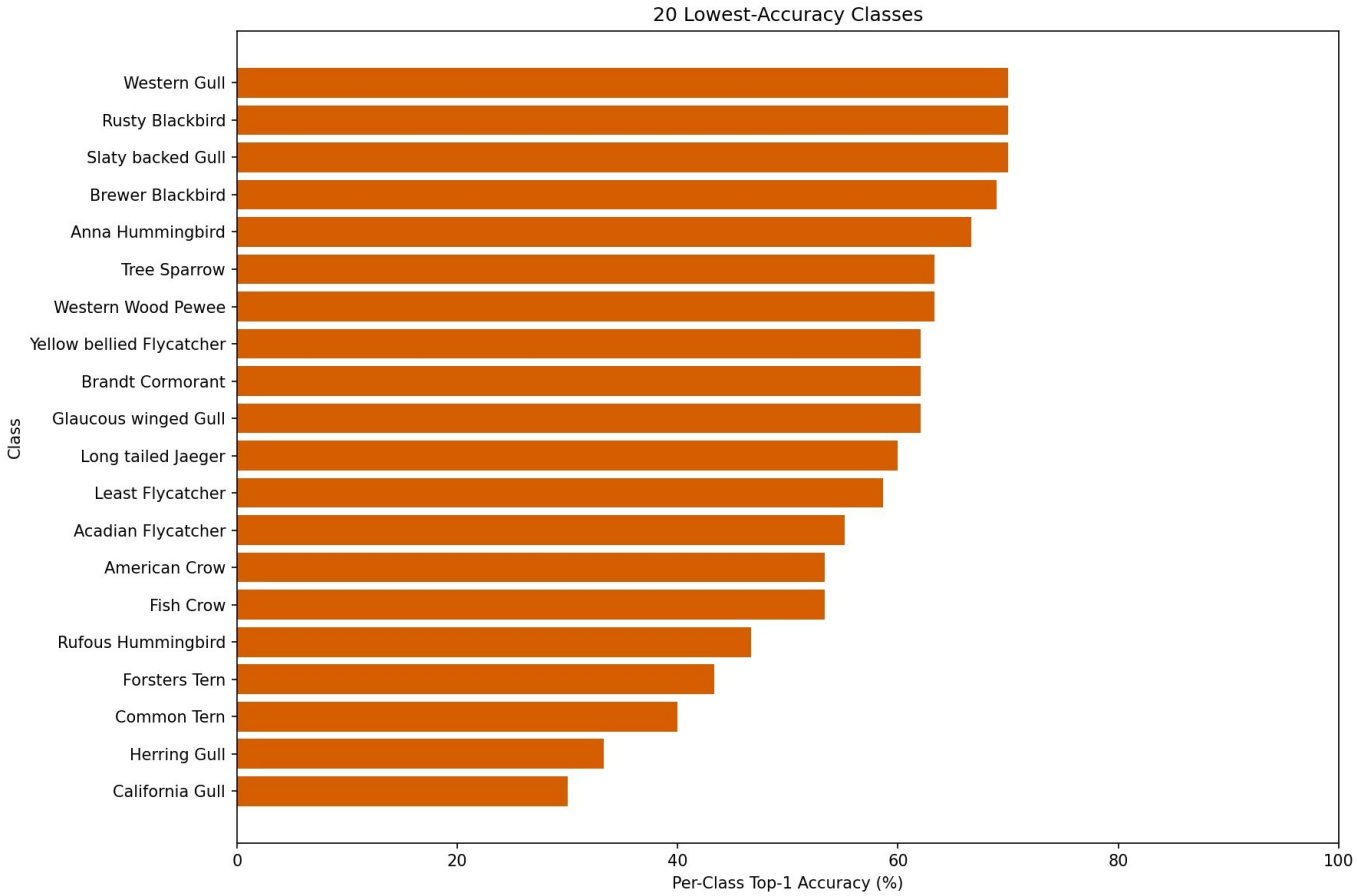

Figure 4. The 20 lowest per-class Top-1 accuracies. California Gull (30%) and Herring Gull (32%) are the hardest classes—both large white gulls that are notoriously difficult even for expert birders. All remaining hard classes correspond to species with high intra-class variation or near-identical congeners (gulls, terns, crows, flycatchers, hummingbirds).

6. Dependencies

Install via `pip install -r requirements.txt`. Core packages: torch 2.2.0, torchvision 0.17.0, transformers 4.36.2, scikit-learn 1.3.2, numpy 1.26.4, matplotlib 3.8.2, seaborn 0.13.0, joblib 1.3.2, Pillow 10.2.0.

7. Limitations & Future Work

The C search range is hand-tuned; a broader logarithmic sweep may find a better optimum. The validation split is deterministic (last 10% per class), so confidence intervals across seeds are unavailable. No test-time augmentation is applied.

Promising directions: (i) DINOv2 with registers for cleaner feature maps; (ii) ViT-S + ViT-B feature concatenation for an ensemble; (iii) bounding-box cropping using CUB-200 part annotations; (iv) replacing logistic regression with a small MLP head.

8. Conclusion

A frozen DINOv2 backbone combined with logistic regression achieves 89.40% Top-1 and 98.36% Top-5 accuracy on CUB-200-2011 in under five minutes of training. The primary failure mode is within visually similar species groups (gulls, terns, corvids), which are near the limits of fine-grained discriminability even for expert observers.

References

[1] Wah, C., et al. *The Caltech-UCSD Birds-200-2011 Dataset.* 2011.

[2] Oquab, M., et al. *DINOv2: Learning Robust Visual Features without Supervision.* TMLR, 2024.

[3] Pedregosa, F., et al. *Scikit-learn: Machine Learning in Python.* JMLR, 2011.

[4] Ranftl, R., et al. *Vision Transformers for Dense Prediction.* ICCV, 2021.